

Last Updated February 26, 2026

Uses PlantUML:

@startuml

title UCSC Financial Purchase Prediction - Component Architecture

skinparam componentStyle rectangle

skinparam shadowing false

```
package "Frontend (React/TS)" {  
    [Dashboard.tsx] as Dashboard  
    [Preview JSON Files\npreview_top_20_data.json\npreview_spend_over_time_data.json] as PreviewJSON  
}
```

```
Dashboard --> PreviewJSON : Preview Mode reads  
}
```

```
package "Backend (FastAPI)" {  
    [main.py\n(FastAPI app + routes)] as MainAPI  
    [analytics.py\n(reads Firestore\ncomputes stats)] as Analytics  
}
```

```
MainAPI --> Analytics : calls functions  
}
```

```
package "Jobs / Pipelines" {  
    [jobs\n(batch tasks)] as Jobs  
    [data_cleaning\n(raw -> clean)] as Cleaning  
    [raw\nExcel/CSV inputs] as RawDir  
    [clean\nCleaned CSV outputs] as CleanDir  
}
```

```
Jobs --> Cleaning : runs cleaning job  
Cleaning --> RawDir : reads  
Cleaning --> CleanDir : writes cleaned CSV  
}
```

```
package "Firebase Upload Layer" {  
    [firebase\n(upload scripts)] as FirebaseScripts  
    [upload_to_firestore.py] as UploadFirestore  
    [upload_to_storage.py] as UploadStorage  
    [test_firestore\n(CSV -> JSON preview)] as TestFirestore  
}
```

```
FirebaseScripts --> UploadFirestore  
FirebaseScripts --> UploadStorage
```

```
Jobs --> FirebaseScripts : triggers upload jobs
TestFirestore --> PreviewJSON : generates
}
```

```
database "Firestore" as Firestore
cloud "Firebase Storage" as Storage
cloud "Google Drive" as Drive
```

```
' ---- Data Source -> Raw ----
Drive --> RawDir : "new Excel added\n(sync/import)"
```

```
' ---- Cleaning -> Firebase ----
UploadFirestore --> Firestore : writes collections\nby upload_id
UploadStorage --> Storage : uploads raw/clean files (optional)
```

```
' ---- Backend reads from Firestore ----
Analytics --> Firestore : queries stats\n(top items, spend over time, etc.)
```

```
' ---- Frontend uses Backend in Live mode ----
Dashboard --> MainAPI : Live Mode fetch\n/api/analytics/*
```

note right of Firestore

Firestore structure (as described):

- upload_ids:
 - amazon
 - cruzbuy
 - pcard

Each upload_id has computed datasets:

- top_items
 - spend_over_time
- (+ others later)

end note

@enduml